

△ **V13 / V14 SCOPE UPDATE — REVISION TO v4.6 IN PROGRESS.** Whitepaper **v4.5** below describes the protocol's **base** architecture as of March 2026 (consensus, monetary policy, four pillars). It predates the 2026-05-18 V13 scope decisions and predates the V13 RC1 release. **For the canonical current scope**, read [docs/V13_PUBLIC_SCOPE_UPDATE.md](#), [docs/V13_GOLD_VAULT_GOVERNANCE_GATES.md](#), [docs/V13_POPC_ESCROW_AUTO_ACTIVATION_GAPS.md](#) and [docs/V13_DTD_FLIP_12100_AUTOMATIC.md](#). The four critical updates relative to v4.5 are:

1. **V13 confirmed at block 12,000** — full cASERT equalizer range **E7-H35** (not the H20 ceiling described in §PILLAR 1 below), DTD lottery cooldown 5 → 6, future-timestamp drift cap 60 s → 30 s (**NTP synchronisation strongly recommended post-V13**), Beacon Phase II-A. RC1 metadata is signed (release_status = signed_metadata_only); binaries upload is the next manual operator step.
2. **V13 verified automatic flip at block 12,100** — the V11 Phase 2 lottery cadence transitions from 2-of-3 bootstrap to 1-of-3 permanent without any operator action, restart, RPC, Beacon notice or config flag. Six audit gates GREEN on main.
3. **V14 fallback at block 15,000** — Gold Vault spend-side governance and PoPC + Escrow automatic lifecycle are **deferred from V13 to V14** (block 15,000, proposed final hardfork in this cycle, not guaranteed). 5 of 9 PoPC consensus gates and 5 of 6 Vault gates are RED today. The accumulation side (25 % per block to both reserves) is unaffected and continues unchanged at consensus since genesis.
4. **Threshold revised: 75 % → 95 % → 90 %** — the Gold Vault miner-signaling supermajority finalised at **≥90 %** over a 67-block window (~12 h, ceil = 61 blocks). The "≥95 %" figure that appears below in §Protocol Summary and §Decentralization reflects the V6-era V14-deferred design and is being rewritten to 90 % in v4.6.

Memory-Lock per-instance, mentioned in earlier drafts as a second anti-pool defense, was **rejected outright** after numerical analysis (penalises small miners proportionally more than large rigs). SbPoW (active since block 7,100) remains the protocol's only anti-pool defense.

// WHITEPAPER

Technical Whitepaper

SOST: Proof of Irreversible Convergence. The complete technical specification covering ConvergenceX consensus, monetary policy, Gold Reserve, PoPC, governance model, security analysis, and constitutional rules. Version 4.5.

v4.5 | March 2026 · No ICO · No Premine · No VCs

The Four Pillars

SOST treats «sound money» as four orthogonal properties that must hold simultaneously and forever. Two of them enforce *credibility of issuance*: the chain advances at a predictable rate (Time) under a measurable, energy-bounded effort (Work). The other two enforce *credibility of value*: total supply is finite (Scarcity) and an explicit fraction is converted, by consensus, into gold-backed reserves (Store of Value). All four are hard-coded at genesis with no governance override. Every section of this whitepaper either specifies one of these pillars or proves it.

▣ PILLAR 1 — TIME

600-second target spacing

Block production is paced at a 10-minute target via the **cASERT unified rate controller**. Difficulty (`bits_q`) adjusts every block from a 288-block running average against target, with bidirectional response: hardens when blocks come too fast, eases when too slow. ConvergenceX equalizer profiles (E1-E7 relief, B0 baseline, H1-H20 hardening pre-V13; **full range H1-H35 at V13, block 12,000**) are mapped from chain lag and protect against runaway acceleration. The E7 floor is immutable; the hardening ceiling rises from H13 (pre-V12) to H20 (V12 from block 7,350) to H35 at V13 activation. From V11 onward a triangular cascade provides emergency relief on stuck blocks without compromising the long-run schedule. See section 3 (cASERT) and section 12.

⚙ PILLAR 2 — WORK

ConvergenceX — native, original

ConvergenceX is a **native, original** Proof-of-Work design. It is **NOT a fork** and **NOT a parameter-tweaked variant** of Bitcoin SHA-256, Monero RandomX, Ethereum Ethash, Equihash, CryptoNight, Scrypt, X11, BLAKE-anything, ProgPoW, RandomHash, or any other prior algorithm. A mining attempt requires solving a 32×32 SPD linear system through 100,000 sequential rounds of integer-only gradient descent and proving the solution sits in a stable basin of attraction (*Proof of Irreversible Convergence*). Memory-hard: **4 GB dataset** (SplitMix64-indexed, state-dependent access from V11 Phase 1) plus **4 GB scratchpad** per mining thread — 8 GB total. Per-block 256-op program changes every block. Verifiers reconstruct sampled rounds with ~500 MB of RAM in ~0.2 ms. See section 3.

◆ PILLAR 3 — SCARCITY

4,669,201 SOST — ever

The hard cap is 4,669,201 SOST — the first seven significant digits of **Feigenbaum's first constant δ** ($\approx 4.6692016091\dots$), the universal ratio that governs the period-doubling route to chaos in nonlinear dynamical systems. The choice is deliberate: ConvergenceX itself is built on the same family of dynamical systems. Emission follows an exponential decay per 131,553-block epoch with quotient $q = e^{-1/4} \approx 0.7788$; epoch 0 reward = 7.85100863 SOST. No premine, no ICO, no dev tax, no minting function, no governance override. See section 4 (Monetary Policy) and Appendix A (Canonical Constants).

◆ PILLAR 4 — STORE OF VALUE

Up to 50% supply → gold-backed

Every block coinbase is split 50 / 25 / 25 by consensus (PoW miner / Gold Funding Vault / PoPC Pool). Over the full emission curve this directs **up to 50% of total supply** into two gold-backed channels:

25% — Perpetual Gold Vault. Protocol-funded, one-way TWAP purchases of tokenised precious metals (XAUT / PAXG today, broader basket later) deposited into a Heritage Reserve on Ethereum mainnet. Observable reserve ratio — *not a peg, not a redemption right*. See section 5.

25% — Proof of Personal Custody (PoPC). Rewards distributed to holders who prove continued custody of gold-backed positions through cryptographic attestation. Models A (self-custody bonds) and B (escrow timelock) collateralise with tokenised gold today and with physically self-custodied gold in the future, preserving operator privacy on-chain. See section 6.

The split is enforced at validation: a block with the wrong coinbase shape is invalid. See sections 5, 6, and Constitution C1-C15 (section 10).

Hardcoded at genesis · immutable forever · no governance override

■

SOST: Proof of Irreversible
Convergence

TECHNICAL SPEC

A mathematically grounded Proof-of-Work based on verifiable dynamical-system certificates. The whitepaper covers the complete protocol specification including consensus algorithm, monetary policy, reserve architecture, custody protocol, and constitutional framework.

Title	SOST: Proof of Irreversible Convergence
Version	v4.5 — March 2026
Authors	NeoB (SOST Protocol)
Format	PDF
Status	Production (mainnet)

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AT A GLANCE

Key Specifications

Protocol Summary

GENESIS CONSTANTS

Name / Symbol	SOST
Genesis	2026-03-15 18:00:00 UTC
Algorithm	ConvergenceX (4GB, 100k rounds, sequential)
Block time	600 seconds target
Max supply	4,669,201.609 SOST (hard cap)
Min unit	1 stock = 0.00000001 SOST
Epoch	131,553 blocks (~2.5 years, Feigenbaum α)
Initial reward	7.85100863 SOST/block
Coinbase split	50% miner / 25% gold / 25% PoPC
Annual decay	~9.03% (smooth, no halvings)
Difficulty	cASERT bitsQ Q16.16, per-block; V5 (block 5,175+): avg288-based, dynamic cap (0%–3.0%); relief valve V8 single E7 cliff at 605 s (block 5,750+) → V9 staged drop 3/60 s from 540 s (block 6,550+) → V10 granular drop 1/60 s from 600 s (block 6,700+); historical: 24h half-life
Reserve	Heritage (sealed by default), Ethereum mainnet
Emergency	≥90% block-based miner signaling over 67-block window (ceil = 61 blocks) + 5-defense Gold Vault model + Transitional Guardian with 10-block pronouncement window and auto-disconnect at block 25,000. Target V14 / block 15,000 (deferred from the original V6 / block 10,000 / ≥95 % design); see V13 scope update.
Premine / ICO	0 — none
Consensus governance	None — immutable at genesis

// CODEBASE FOOTPRINT

Codebase vs Bitcoin Core

Lines-of-code totals for the SOST stack as of **2026-05-10**, side by side with the Bitcoin Core reference implementation. LOC is a rough engineering metric — **not** a quality, security or decentralisation metric — but it gives a sense of how much surface area the project covers.

Component	SOST	Bitcoin Core
Blockchain core (C++ / protocol)	36,442	~280-310k
Front-end & explorer (HTML + JS + CSS)	69,975	—
Materials Engine (Python)	161,282	—
Geaspirit — geo-mineral ML (Python)	9,386	—
Auth gateway (Python)	704	—
Documentation (markdown)	23,975	(separate)
Total code (excl. docs)	~278,000	~280-310k
Total code + docs	~302,000	—

1. The blockchain core alone is 5-7× smaller than Bitcoin Core. Bitcoin Core has 15+ years of P2P hardening, scripting (Script + Taproot), thousands of tests, and dozens of network conditions handled in production. SOST does not, and pretending otherwise would be dishonest.

2. The full SOST stack matches Bitcoin Core in order of magnitude only because SOST is a vertical. The total includes a Materials Discovery Engine (161k LOC) and a geological prospectivity ML stack (Geaspirit) that Bitcoin Core does not contain. Comparing totals across different problem domains is not apples-to-apples.

3. What the numbers do show. SOST is a serious open-source project (~300k LOC total) with depth beyond a typical altcoin fork — the consensus is custom (ConvergenceX + cASERT), the wallet model is in-house, the front-end is hand-rolled, and the protocol is integrated with two external scientific systems.

4. What the numbers do NOT show. Network effect, peer count, security audits, deployed value, decentralisation. Those are independent of LOC and Bitcoin leads on every one of them.

Counts measured 2026-05-10 with `find ... | xargs wc -l` over each component's own source tree, excluding build artefacts and vendored code. Bitcoin Core figure is the upstream bitcoin/bitcoin master snapshot, code + tests, no docs.

// TRADING

DEX and OTC P2P — simple & secure

SOST ships two trading surfaces. Both are **non-custodial**, both are **simple by design**, and both have an explicit security model. Fiat trades are **not offered as a protocol feature** — the SOST chain cannot verify a bank, so the protocol refuses to pretend it can.

SOST DEX (PoPC) — already secure today

The SOST DEX trades PoPC contracts (Proof of Personal Custody positions backed by tokenised or physical gold). Every trade combines three independent security layers:

- **Client-side ED25519-signed offers.** The offer payload (action, position-id, maker SOST address, maker ETH address, SOST amount, gold amount, price, expiry) is signed in the user's browser. The signing key is encrypted with Argon2id from the user's passphrase and never leaves the device.
- **End-to-end encrypted relay.** X25519 envelopes carry the signed offers between users. The relay server transports bytes but cannot read the cleartext, cannot alter the payload (MAC verified client-side), and cannot impersonate either party.
- **On-chain Ethereum escrow.** Where gold tokens move (XAUT, PAXG), they sit in the audited `SOSTEscrow.sol` smart contract. **SOST Protocol holds no keys to that contract.** Releases happen by on-chain logic the two parties trigger themselves — no admin, no operator, no oracle.

The protocol cannot reverse trades, cannot recover funds sent to a wrong address, cannot compel delivery, and cannot verify off-chain promises. **That is the design.** What it does instead: enforces structural integrity of every signed payload, anchors every trade in cryptographic primitives, and surfaces 26 known scam patterns via the client-side Sentinel chat layer. The user's responsibility is to (a) load only the official URL, (b) read the signed fields before signing, (c) verify counterparty addresses against the payload, (d) check txids on-chain, and (e) use small test transactions before larger amounts.

SOST OTC / P2P Board — crypto-to-crypto via atomic swap

The community OTC / P2P Board hosts user-to-user trades against seven counterparty assets: **BTC, ETH, USDT, USDC, BNB, PAXG, XAUT**. The intended settlement mechanism is the **atomic swap HTLC** (Hashed Time-Locked Contract): cryptography enforces that **either both parties receive their funds, or both refund** — no intermediate state exists in which one side walks away with everything. No SOST escrow, no admin custody, no first-mover risk, no protocol authority.

The implementation status (as of this revision) is:

- **SOST-side consensus** — **code-complete and unit-tested**. HTLC_LOCK (R17), HTLC_CLAIM (R19-R22), HTLC_REFUND (R23-R24) validation rules are in `src/tx_validation.cpp`, gated by `ATOMIC_SWAP_HTLC_ACTIVATION_HEIGHT`.
- **Wallet, RPC, and CLI** — **scaffolding complete**. `createhtlclock`, `claimhtlc`, `refundhtlc`, `decodehtlc`, `gethtlcstatus` exist as gated helpers.
- **Counterparty legs (BTC Script, EVM contract)** — **pending Phase 4**. Required before any swap can settle.
- **External cryptographic + economic review** — **pending Phase 5**. Required before the gate flips.
- **Activation** — the gate stays at `INT64_MAX` (sentinel OFF). Target flip: **V14 / block 15,000**, conditional on Phase 4 + 5 sign-off.

The trust profile of the seven assets splits into two categories. **Trust-minimized (BTC, ETH, BNB)**: no token-issuer freeze risk; cryptographic atomicity holds on both sides. **Issuer-risk (USDT, USDC, PAXG, XAUT)**: the underlying token can be frozen by its issuer (Tether, Circle, Paxos, TG Commodities); if that happens mid-swap, the SOST side still refunds correctly but the counterparty side may become uncollectible. The DEX UI labels these four explicitly and recommends small amounts for them; larger amounts should stick to Category A.

OTC against fiat — deliberately not offered as a protocol feature

A trade where one leg is a SEPA transfer, wire, or cash payment **cannot be made cryptographically atomic**. The SOST chain has no view into a bank. Any mechanism that would "penalise the unpaid fiat side automatically" requires either (a) an oracle that reads the bank — which is arbitration and turns SOST into a regulated intermediary — or (b) one party self-declaring incumplimiento, which is a trivially-exploitable robbery primitive; both also enable Sybil collusion farms. There is no third path.

Therefore **SOST Protocol does not offer fiat OTC as a protocol service**. The community discussion board still permits conversations about fiat trades (people are free to do whatever they want with each other), but the protocol provides no escrow, no penalty, no dispute resolution, and no implied safety. **The protocol's strong recommendation is to keep all trades in the crypto-to-crypto path** (atomic

swap) where the cryptography itself can enforce safety. Any fiat trade is off-protocol, at user risk, with no protocol safety net.

The simplicity rule

Both surfaces follow the same five-step UI discipline so the user can be safe without becoming a cryptographer:

1. Load only the official URL: `sostcore.com` or `sostprotocol.com`. Never click a link from a DM.
2. Read the signed offer fields before signing. Counterparty address, amount, expiry — all visible, all binding.
3. Verify counterparty addresses against the LOCK / escrow payload. The signed payload is authoritative; chat messages are not.
4. Confirm txids on-chain (SOST explorer or Etherscan) before releasing anything. Screenshots are not proof.
5. Do a small test transaction before any larger amount. Always. Without exception.

Anything sold outside this flow — Telegram DMs, "official escrow admin", address changes by chat, advance fees, screenshot proofs — is a scam pattern. The client-side Sentinel flags 26 known variants at the chat layer. The OTC and DEX pages document each pattern with worked examples. **If a counterparty pushes you off the official flow, stop the trade.**

// DECENTRALIZATION

Honest Assessment

The network and mining layers are already partially decentralized: multiple independent nodes, multiple active miners, open-source code, and no admin keys for block production.

However, two economic custody components are still temporarily centralized during Phase I (until governance activates — originally targeted for V6 / block 10,000, deferred

to **V14 / block 15,000** after the 2026-05-18 scope review; see banner above and docs/V13_PUBLIC_SCOPE_UPDATE.md):

- **Gold Vault** (25% of every block reward): developer-held custody until the 5-defense governance model activates — **V14 / block 15,000** target (deferred from V6 / block 10,000 after the V13 review; 5 of 6 governance gates are RED today, full audit at docs/V13_GOLD_VAULT_GOVERNANCE_GATES.md)
- **PoPC Pool** (25% of every block reward): developer-held custody until PoPC + Escrow operational launch — **V14 / block 15,000** target (5 of 9 PoPC consensus gates are RED today, full audit at docs/V13_POPC_ESCROW_AUTO_ACTIVATION_GAPS.md)

The protocol developer cannot change the constitutional 50/25/25 split, because it is hardcoded at genesis. What remains centralized during Phase I is *custody*, not the emission rule itself.

Custody transition roadmap:

- **Block 15,000 (V14 fallback hard fork):** Gold Vault governance enforced at consensus level — 5-defense model (purpose restriction, dual destination whitelists, per-spend cap, rate limiting, **≥90 % block-based miner signaling over a 67-block window**) plus the Transitional Guardian role (authorise / veto with 10-block pronouncement window; silence ⇒ miner result prevails; auto-disconnect at block 25,000, implemented in V13 source from day one, no future fork required). The protocol developer loses executive vault control. PoPC + Escrow operational launch under the same fork. Updated threshold history: 75 % (original whitepaper) → 95 % (V6 internal review) → 90 % (V13 review, finalised after observing real network behaviour).
- Gold reserve purchases become externally verifiable on Ethereum (already started: 1.2 oz committed)
- **Phase III (12-24 months):** Migration from tokenized gold (XAUT/PAXG) to physical gold custody in a regulated vault. No bridges, no wSOST, no cross-chain attack surface.

The custody arrangement during Phase I is at the protocol developer's operational discretion. The transition to protocol custody at block 10,000 has a hard date encoded in the consensus rules — it is not subject to the developer's later willingness to step back. *The Gold Vault must be able to change form, but not purpose.*

// POSITIONING

SOST vs other PoW chains

SOST is not positioned as a Bitcoin clone. It combines Bitcoin-like timing discipline, CPU-heavy original work, mathematical scarcity, and a constitutional reserve design. The four tables below place SOST against six well-known PoW chains so the reader can see exactly where the design overlaps and where it diverges.

Snapshot below is **historical** (frozen 2026-05-08 UTC) and is not refreshed on every page load. Block heights and schedule drift change every minute; **see the live explorer at sost-explorer.html** for the current SOST height. Sources for the comparison columns: Blockchain.info (BTC), BlockCypher (LTC, DOGE), localmonero.co (XMR), Blockchair (BCH, ZEC).

1. Schedule — ahead, behind, or on time

Chain	Target / block	Height now	Expected	Drift	Status
Bitcoin	600 s (10 min)	948,511	~912,200	+36,300	Ahead ~250 days
Litecoin	150 s (2.5 min)	3,103,448	~3,063,000	+40,400	Ahead ~70 days
Dogecoin	60 s (1 min)	6,198,544	~6,531,800	-333,300	Behind ~232 days
Monero	120 s (60 s before 2016)	3,669,797	~3,672,000	-2,200	Essentially on time
Bitcoin Cash	600 s	950,193	~938,800	+11,400	Ahead ~80 days
Zcash	75 s (150 s pre-Blossom)	3,335,642	~3,349,000	-13,000	Essentially on time
SOST	600 s	7,713	~7,700	+11...13	Essentially on time

Reading: chains with bulk retarget (BTC, LTC) accumulate large positive drift because their hashrate has grown exponentially faster than two-week recalibration can correct. BCH switched

to ASERT in 2020 and the drift growth flattened. Dogecoin’s drift is negative because its merge-mining economy with Litecoin and DigiShield retarget combine to produce blocks slower than its 1-minute target. Monero, Zcash and SOST track their schedule within a fraction of a day — the property delivered by per-block retarget.

2. Emission and supply

Chain	Block reward (now)	Next change	Final supply	Notes
Bitcoin	3.125 BTC	2028 → 1.5625	21,000,000 BTC	Halvings every 210k blocks (~4 yr)
Litecoin	6.25 LTC	~2027 → 3.125	84,000,000 LTC	Halvings every 840k blocks (~4 yr)
Dogecoin	10,000 DOGE (fixed)	never	No cap	Perpetual inflation, ~5.26 B DOGE / yr
Monero	~0.6 XMR (tail)	fixed since 2022	~18.4 M + tail	Tail emission 0.6 XMR/block forever from June 2022
Bitcoin Cash	3.125 BCH	2028 → 1.5625	21,000,000 BCH	Inherited BTC’s halving calendar at the 2017 fork
Zcash	1.5625 ZEC	2028 → 0.78125	21,000,000 ZEC	Halvings every 4 yr
SOST	7.85100863 SOST (epoch 0)	block 131,553 (~2.5 yr)	4,669,201 SOST	Continuous decay $q = e^{-1/4} \approx 0.7788$ per epoch

Bitcoin and its derivatives use **discrete halvings**. Monero and SOST use **continuous decay**. Doge does not decay at all (perpetual inflation). Monero’s tail emission means its supply has no asymptote — only a curve that flattens to a constant absolute issuance.

Chain	PoW algorithm	Dominant hardware	Retarget	Notable design
Bitcoin	SHA-256d (double SHA-256)	ASIC	Bulk every 2,016 blocks (~14 d)	The original. Stateless flat hash.
Litecoin	Scrypt	ASIC	Bulk every 2,016 blocks (~3.5 d)	Scrypt holds a 128 KB working buffer; ASIC-friendly since 2014.
Dogecoin	Scrypt (merge-mineable with LTC)	ASIC via LTC	DigiShield (per-block) since Feb 2014	Security tied to Litecoin's merge-mined hashrate.
Monero	RandomX	CPU (deliberate)	LWMA (per-block, 720-block window)	CPU-oriented VM with JIT execution of randomised bytecode programs and a multi-MB-class dataset/cache plus per-execution scratchpads. ASIC-hostile by construction.
Bitcoin Cash	SHA-256d (same as BTC)	ASIC	ASERT (per-block) since Nov 2020	Per-block since the post-fork DAA bug of 2017.
Zcash	Equihash (200, 9)	ASIC (since 2018)	DigiShield-like (per-block, EMA)	Originally GPU-friendly memory-hard; ASICs reached the chain by mid-2018.
SOST	ConvergenceX	CPU + 8 GB RAM	cASERT (per-block, avg288 + E/B/H equalizer profiles)	Native, original algorithm — not a fork. Solves a 32×32 SPD linear system across 100k sequential rounds of integer-only gradient descent and proves the solution sits in a stable basin (<i>Proof of Irreversible Convergence</i>). 256-op

Chain	PoW algorithm	Dominant hardware	Retarget	Notable design
				per-block program. 4 GB dataset + 4 GB scratchpad per mining thread.

4. Properties at a glance

Property	BTC	LTC	DOGE	XMR	BCH	ZEC	SOST
Hard cap	✓ 21M	✓ 84M	✗ infinite	✗ tail	✓ 21M	✓ 21M	✓ 4.67M
Per-block retarget	✗ bulk	✗ bulk	✓ DigiShield	✓ LWMA	✓ ASERT	✓ DigiShield	✓ cASERT
CPU-oriented PoW	✗	✗	✗	✓ RandomX	✗	✗ (was GPU)	✓ ConvergenceX
Memory-hard footprint	✗	~ 128 KB	✗	✓ RandomX (multi-MB class)	✗	~ partial	✓ 8 GB total
Native (not a fork)	✓	✗	✗	~ CryptoNight lineage	✗ BTC fork	✗ Equihash	✓ ConvergenceX original
Constitutional reserve	✗	✗	✗	✗	✗	✗	✓ 50% gold-linked
Default-private transactions	✗	✗	✗	✓ ring + stealth	✗	~ shielded opt-in	✗ (E2E layer in DEX/Talk)
Hard cap derived from a math constant	✗	✗	n/a	n/a	✗	✗	✓ Feigenbaum δ

Where SOST stands out among the chains in this table

1. The only PoW algorithm in this table that is genuinely native and not derived. SHA-256d was Bitcoin's original; Scrypt comes from Tarsnap; Equihash from Wagner's generalised birthday. Monero's RandomX is the closest other example of an originally designed algorithm, but it carries CryptoNight ancestry. ConvergenceX shares no core structure with any of these.
2. The only hard cap in this table derived from a universal mathematical constant (Feigenbaum's δ). The other caps are round numbers chosen for marketing alignment.
3. The only chain in this table with a **protocol-level allocation toward gold-linked reserves and custody mechanisms**: 25% Perpetual Gold Vault one-way TWAP purchases, plus 25% PoPC. *This is not a peg, not a redemption claim, not a stablecoin mechanic.* The reserve ratio is observable on-chain, not contracted.
4. Among the chains in this table, SOST combines per-block retarget with the largest mining-RAM footprint (8 GB) of any production CPU-oriented design. Monero is the only direct competitor on the CPU-friendly axis but uses a smaller working set.
5. Tightest schedule alignment in the comparison: SOST is currently within ~13 blocks of its target, helped by cASERT's 288-block average and dynamic delta cap (0% / 0.5% / 1.5% / 2.5% / 3% by deviation).
6. Hard cap lower than the major PoW comparables in this table. With 4.67M coins capped, SOST's long-run scarcity is steeper than BTC, LTC, BCH, or ZEC. Whether that matters in market terms is a separate question.

Where SOST does not stand out — honestly

- **Default privacy:** Monero wins on transaction privacy by default. SOST has end-to-end encryption in DEX/Talk but the base ledger is transparent like Bitcoin.
- **Track record / hashrate / market depth:** SOST is days-months old; BTC is 17 years old. Comparing security-economic depth today would be misleading.
- **Liquidity and exchange listings:** SOST is pre-market — public launch phase opens 2026-06-16, exchange outreach follows from there. The chains in this table all have active markets.
- **External security audit:** none yet. Independent reviewers welcome.
- **Post-quantum signature safety:** the current scheme uses secp256k1 + BIP-340 Schnorr and is not PQ-resistant. PQ migration is on the roadmap as a future hard fork.

Comparison block last updated alongside the snapshot above. Tables refresh on each whitepaper revision.

SOST's first **scientific vertical**

Trinity is a vertical built *on top* of the SOST chain. The chain stays a strict Bitcoin-class PoW ledger; Trinity adds a **discovery + proof + reputation** layer above it. Two tracks run today: **Earth Track** (GeaSpirit-driven AOI proposals) and **Materials Track** (autonomous oxide candidate proposals). Both produce hash-bound proof bundles whose root SHA can be anchored on chain as a single capsule. The first anchor — Kalgoorlie Phase 1 at block #8085, txid d68678b5... — is the pipeline's public birth certificate.

Six-stage pipeline

1. **Generate hypotheses.** Deterministic seed → candidate AOIs or candidate compositions.
2. **Filter.** Charge balance, toxicity gate, criticality flag, known-demo exclusion, protected-area gate.
3. **Review.** SOST AI Council (free-tier: validator + local_knowledge + mock_ai) decides hold / accept / reject / contradicted with validator-veto, agreement / disagreement summary and confidence. Deterministic, offline, no paid calls. From v0.2 onward, the same canonical `multi_ai_review.AICouncil` reviews both Earth Track and Materials Track hypotheses; no per-track mock remains.
4. **Plan heavy work.** Propose DFT, MLIP, phonon screening, satellite reprocessing or geophysics fusion. Dry-run; no rewards.
5. **Package.** Scorecard + dossier + plan + campaign manifest → one canonical proof bundle JSON + Markdown sidecar with Merkle root over four SHA-256 anchors.
6. **Seal on SOST.** Operator-driven manual capsule registration. One TXID, one block height, one immutable timestamp. Not automated; not consensus-affecting.

What this chain proves — and what it does not

SOST does not certify that a candidate material is revolutionary, that a candidate AOI hosts a deposit, or that any laboratory or field result has been obtained. Trinity bundles state the opposite explicitly: *not experimentally validated, not DFT validated, not a patent claim, not a commercial performance claim*. What SOST proves is something narrower but durable: **this exact reproducible scientific process produced these exact documents at this exact block height, and they have not been altered since**. That priority + traceability gives the work a public, immutable timestamp, which is the precondition for any future reputation, ranking or reward system the protocol may host.

Six operator-enforced rules

1. Trinity never touches consensus, miner code, node code, RPC schema or the wire format.
2. Trinity never promises validated discoveries; every artefact carries explicit honesty markers.
3. Trinity always produces hash-bound, offline-verifiable bundles.
4. Trinity registers only milestones, not every iteration; on-chain anchoring is an operator-driven manual event.
5. Useful Compute rewards stay off until a separate governance event activates them.
6. Public framing is utility-first: the engines produce scientific intelligence; SOST is the trust layer around that work.

One-line summary. Trinity is SOST's first strong use case: *turning autonomous digital research into publicly verifiable proof.*

Full Trinity reference: sost-trinity.html · live proof registry on the public mirror.
